

Chapter 2 – System Requirements & Download

Learning Objectives

After completing this chapter, you will be able to:

- Understand the minimum and recommended hardware requirements for Kali Linux.
 - Choose the correct Kali Linux installation image.
 - Download Kali Linux from the official website.
 - Verify the downloaded ISO using SHA256 checksums.
 - Download Oracle VirtualBox.
 - Install the VirtualBox Extension Pack.
 - Prepare your system for creating a Kali Linux virtual machine.
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2.1 Why Do We Need System Requirements?

Before installing any operating system, it is important to verify that your computer has sufficient hardware resources.

Meeting the recommended requirements helps ensure that Kali Linux runs smoothly, especially when using resource-intensive security tools like Burp Suite, Wireshark, Metasploit, and Ghidra.

2.2 Minimum Hardware Requirements

These specifications are suitable for basic learning and lightweight labs.

Component	Minimum Requirement
CPU	2 Core Processor
RAM	2 GB

Storage	20 GB
Virtualization	Intel VT-x / AMD-V Enabled
Internet	Required for Updates

Note: Kali Linux can run with lower specifications, but performance may be limited.

2.3 Recommended Hardware Requirements

For a better learning experience and cybersecurity home labs, use the following configuration.

Component	Recommended
CPU	4 Cores or More
RAM	4–8 GB
Storage	50 GB or More
Virtualization	Enabled
Internet	Stable Connection

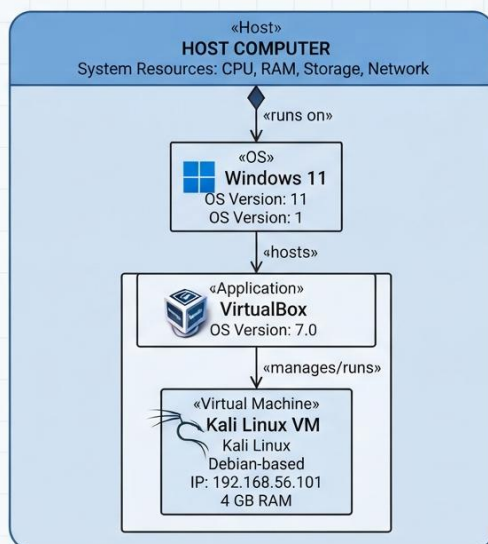
This configuration allows you to comfortably run multiple tools and virtual machines.

2.4 Host Machine Requirements

Your host computer should have enough resources to run both Windows (or another host OS) and Kali Linux simultaneously.

Example:

UML DIAGRAM: HOST COMPUTER & VIRTUALIZATION



If your computer has 8 GB RAM, avoid allocating all of it to the virtual machine. Leave enough memory for the host operating system.

2.5 Virtualization Support

Virtualization allows you to run one operating system inside another using software like Oracle VirtualBox.

Modern processors include hardware virtualization technologies:

- Intel VT-x
- AMD-V

These features improve the performance of virtual machines.

How to Check Virtualization

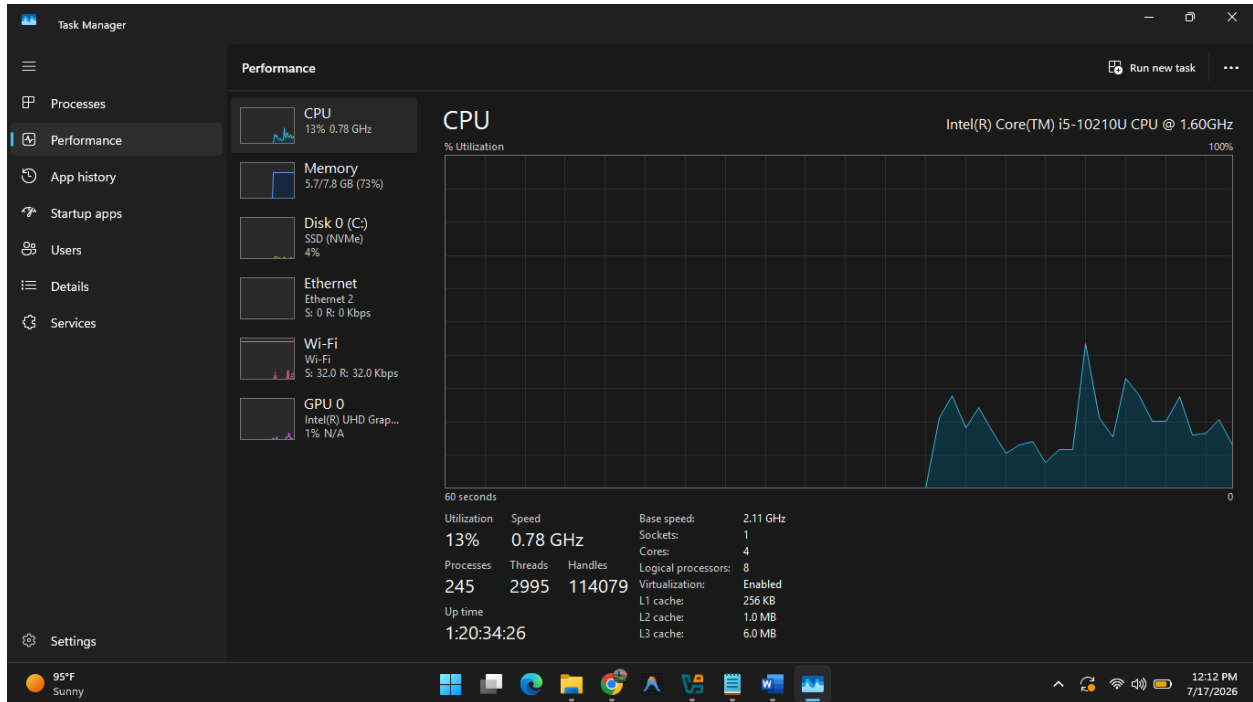
Windows Task Manager

1. Press **Ctrl + Shift + Esc**
2. Open the **Performance** tab.

3. Select **CPU**.
4. Check **Virtualization**.

Example:

Virtualization: Enabled



2.6 What Is an ISO File?

An **ISO image** is a complete copy of an operating system installation disc.

Instead of installing from a DVD or USB drive, VirtualBox boots directly from the ISO file.

Kali ISO

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Virtual Machine

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Install Kali Linux

Think of an ISO as a digital installation DVD.

2.7 Choosing the Correct Kali Linux Image

The official Kali website provides several download options.

Image	Purpose
Installer ISO	Install Kali on a physical or virtual machine
Live ISO	Run Kali without installation
Virtual Machine	Pre-built image for VirtualBox or VMware
ARM	Raspberry Pi and ARM devices
Cloud	Cloud platforms

For this guide, we will use the **Installer ISO**.

2.8 Download Kali Linux

Always download Kali Linux from the official website.

Official Website:

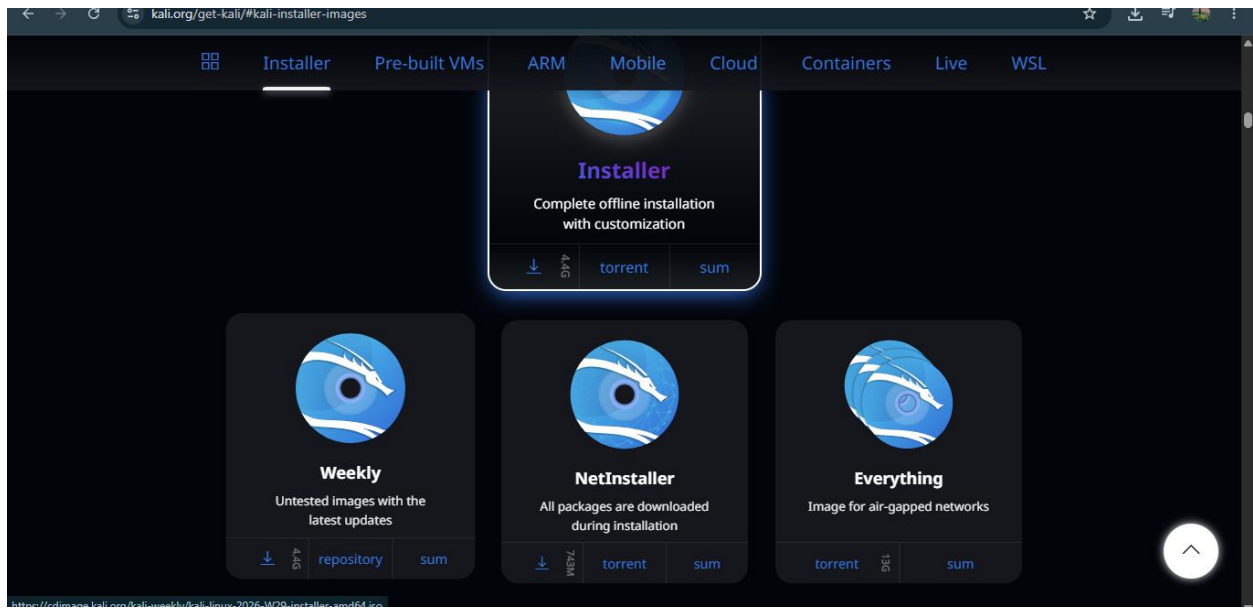
<https://www.kali.org/get-kali/>

Choose:

Installer Images

Then download:

64-bit Installer ISO



2.9 Why Download from the Official Website?

Downloading from unofficial sources can expose you to:

- Malware
- Modified ISOs
- Outdated versions
- Security risks

Always verify that the download comes from:

<https://www.kali.org>

2.10 Verifying the ISO (SHA256)

After downloading the ISO, verify its integrity using the published SHA256 checksum.

This ensures:

- The file downloaded completely.
- The file has not been modified.

- The ISO is authentic.

Why Is Verification Important?

Imagine downloading a 4 GB ISO.

If the download is interrupted or corrupted, the installation may fail.

SHA256 helps confirm that your downloaded file matches the original.

2.11 Download Oracle VirtualBox

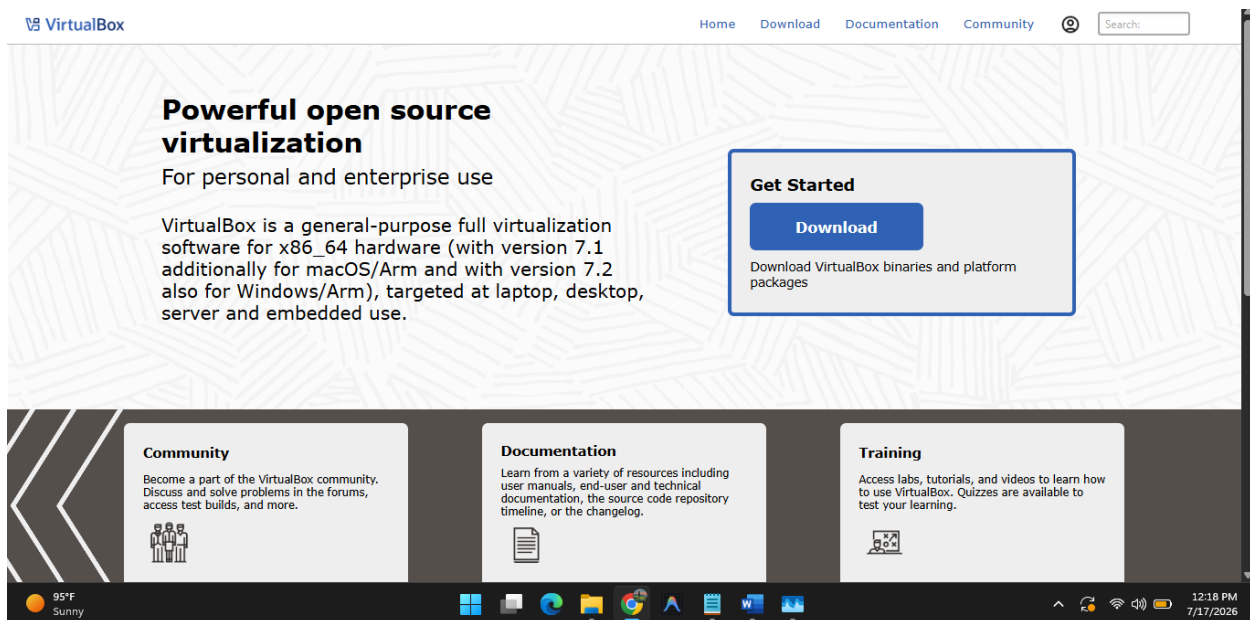
Oracle VirtualBox is free virtualization software that allows multiple operating systems to run simultaneously.

Official Website:

<https://www.virtualbox.org/>

Download:

Windows Hosts



2.12 Install Oracle VirtualBox

Run the installer and follow the setup wizard.

Recommended installation:

Next→Next→Next→Install→Finish

Keep the default settings unless you have a specific reason to change them.

2.13 Download the VirtualBox Extension Pack

The Extension Pack provides additional features, including:

- USB 2.0 and USB 3.0 support
- Virtual webcam support
- Disk encryption
- PXE boot for Intel network cards
- Other advanced virtualization capabilities

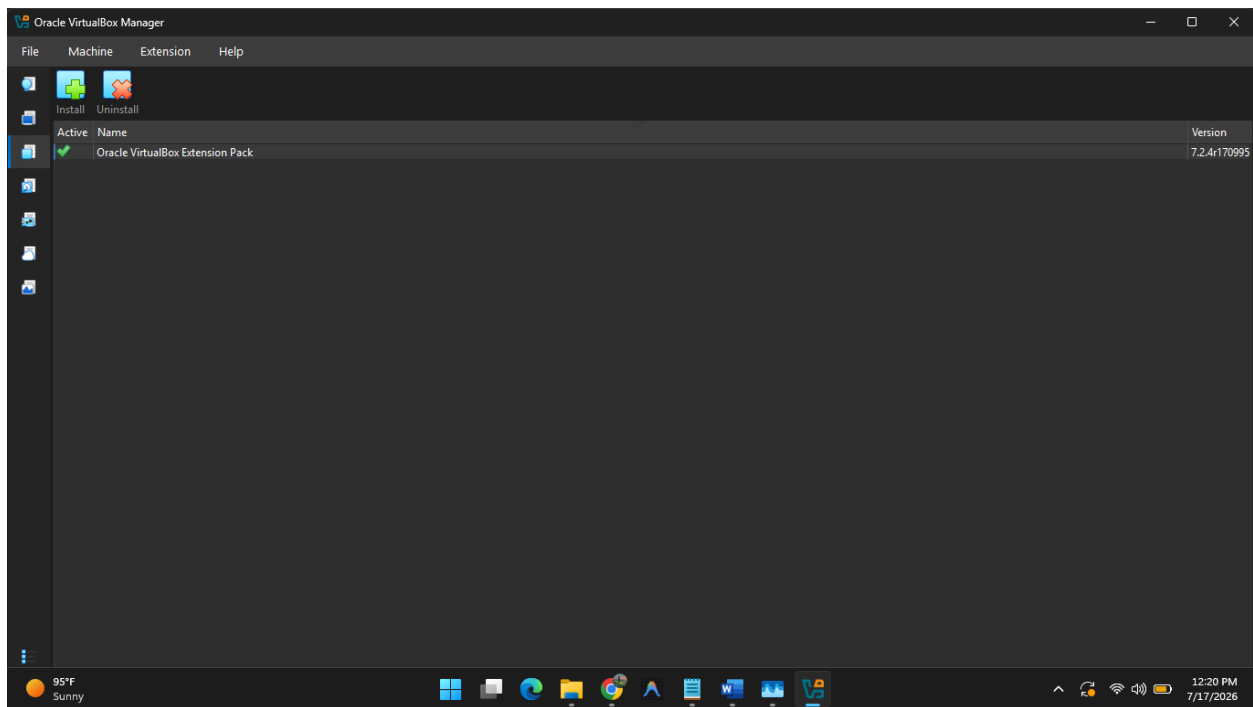
Download it from:

<https://www.virtualbox.org/wiki/Downloads>

2.14 Installing the Extension Pack

After downloading:

1. Open VirtualBox.
2. Go to **File** → **Tools** → **Extension Pack Manager** (or **Tools** → **Extensions**, depending on your VirtualBox version).
3. Click **Install**.
4. Select the downloaded .vbox-extpack file.
5. Accept the license agreement.
6. Complete the installation.



2.15 Folder Organization (Recommended)

Create a dedicated folder for your lab files.

Example:

Cybersecurity-Lab/

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|— ISOs/

| |— kali-linux.iso

| |— ubuntu-server.iso

|

|— Virtual Machines/

|

|— Tools/

|

└─ Backups/

Keeping your files organized makes future maintenance and backups much easier.

Best Practices

- Download software only from official websites.
 - Verify ISO files using SHA256 before installation.
 - Ensure virtualization is enabled in your BIOS/UEFI.
 - Keep at least 20–30 GB of free disk space available.
 - Store ISO files in a dedicated folder for future use.
 - Install the VirtualBox Extension Pack to access additional virtualization features.
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Chapter Summary

In this chapter, you learned:

- The minimum and recommended hardware requirements for Kali Linux.
 - The purpose of virtualization and hardware virtualization support.
 - What an ISO file is and why it is used.
 - How to download the official Kali Linux Installer ISO.
 - Why verifying the ISO with SHA256 is important.
 - How to download and install Oracle VirtualBox.
 - How to install the VirtualBox Extension Pack.
 - How to organize your lab files for a clean and maintainable environment.
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One-Line Summary

Preparing the correct hardware, downloading trusted installation files, verifying their integrity, and installing Oracle VirtualBox are essential first steps for building a reliable Kali Linux virtual machine.
